

Power BI Business Intelligence Project

Sales Performance, Profitability & Business Intelligence Dashboard

1. Business Background

A retail organization operates across multiple locations and sells a wide range of products through a team of salespeople. The organization has accumulated sales data for **2021, 2022, and 2023**, along with information about products, customers, locations, salespeople, and monthly budgets.

Currently, business managers rely on manually prepared Excel reports to understand sales performance. This makes it difficult to:

- Track overall sales performance
- Compare performance across years
- Identify high-performing products and locations
- Evaluate salesperson performance
- Understand customer contribution
- Compare actual sales against budget
- Analyze sales trends over time
- Identify areas of business growth or concern
- Make data-driven decisions quickly

The management wants a **Power BI Business Intelligence Dashboard** that provides a centralized view of the organization's sales performance.

2. Business Problem Statement

You are working as a **Business Intelligence Analyst** for the organization.

Your responsibility is to transform the available sales and business data into an interactive Power BI solution that helps management answer:

"How is our business performing, where are we generating the most revenue and profit, what factors are driving performance, and where should management focus its attention?"

You are expected to import, clean, transform, model, analyze, and visualize the data using **Microsoft Power BI**.

The final dashboard should allow business users to interact with the data and drill down from overall business performance to product, customer, salesperson, location, and time-level insights.

3. Dataset Provided

The Excel workbook contains the following tables:

Table	Purpose
2021 Sales	Sales transactions for 2021
2022 Sales	Sales transactions for 2022
2023 Sales	Sales transactions for 2023
Products	Product master information
Locations	Location and geographical information
Customers	Customer information
Sales People	Salesperson information
Budgeting	Budget/revenue allocation information
Date Ranges	Time-period information for analysis

4. Your Objective

Build a **professional Power BI Sales Intelligence Dashboard** that provides management with:

1. Overall business performance
2. Year-over-year sales analysis
3. Product performance
4. Customer performance
5. Salesperson performance
6. Geographic performance
7. Profitability analysis
8. Budget vs Actual analysis
9. Time-based sales trends

10. Interactive business insights

5. Phase 1 – Data Understanding & Data Preparation

Before building visuals, understand the structure and quality of the data.

Tasks

- Import all relevant sheets into Power BI.
- Inspect every table.
- Identify primary keys and possible foreign keys.
- Check data types.
- Check for null/blank values.
- Identify unnecessary columns.
- Identify duplicate records.
- Standardize column names where required.
- Check whether the three yearly sales tables have consistent structures.
- Append the 2021, 2022 and 2023 sales tables into a single Sales table.
- Rename the final table appropriately, for example:

Fact_Sales

Power Query Requirements

Students should demonstrate:

- Data type transformation
- Column renaming
- Removing unnecessary columns
- Handling blank/null values
- Appending tables
- Merging tables where appropriate
- Creating calculated columns where required
- Data profiling

6. Phase 2 – Data Modeling

Create an appropriate **Power BI data model**.

Students should identify:

Fact Table

Fact_Sales

Possible fields:

- Order ID
- Product ID
- Location ID
- Sales Person ID
- Customer ID
- Order Date
- Quantity
- Price

Dimension Tables

- Dim_Product
- Dim_Location
- Dim_Customer
- Dim_Salesperson
- Dim_Date

Students should determine the appropriate relationships between these tables.

Expected Modeling Concept

Students should aim for a **Star Schema** rather than connecting every table directly to every other table.

Example:

Dim_Product → Fact_Sales ← Dim_Customer

Dim_Location → Fact_Sales

Dim_Salesperson → Fact_Sales

Dim_Date → Fact_Sales

7. Phase 3 – Date Table

Create a proper Date dimension.

The Date table should support:

- Year
- Quarter
- Month
- Month Number
- Month Name
- Week
- Day
- Year-Month

You should ensure that months are displayed in chronological order rather than alphabetical order.

The Date table should be marked as the official **Date Table** in Power BI.

8. Phase 4 – Create Business Measures Using DAX

You should create appropriate DAX measures instead of relying only on implicit aggregation.

At minimum, calculate:

Sales Metrics

- Total Sales
- Total Orders
- Total Quantity Sold
- Average Order Value
- Average Selling Price

Cost & Profitability

Using the product cost information, calculate:

- Total Cost
- Gross Profit
- Gross Margin %
- Revenue per Unit
- Profit per Unit

Time Intelligence

Calculate:

- Current Year Sales
- Previous Year Sales
- YoY Sales Growth
- YoY Growth %
- MTD Sales
- YTD Sales
- Previous Year YTD Sales

Budget Analysis

Calculate:

- Budget
- Actual Sales
- Budget Variance
- Budget Achievement %
- Actual vs Budget

you should clearly understand the difference between:

Calculated Column vs Measure

and use each appropriately.

9. Phase 5 – Executive Dashboard

Create an **Executive Sales Dashboard** for senior management.

The dashboard should answer:

"How is the business performing?"

Include KPI cards such as:

- Total Sales
- Total Profit
- Profit Margin %
- Total Orders
- Total Quantity
- Average Order Value
- Budget Achievement %
- YoY Growth %

Recommended Visuals

- KPI Cards
- Sales Trend Line Chart
- Sales by Year
- Sales by State
- Top 10 Products
- Top 10 Customers
- Top Salespeople
- Actual vs Budget

Management should be able to quickly understand the overall health of the business from this page.

10. Phase 6 – Sales Trend Analysis

Create a dedicated **Sales Trend Analysis** page.

Analyze:

- Daily sales
- Monthly sales
- Quarterly sales

- Yearly sales
- Year-over-year growth
- Seasonal patterns

Business Questions

1. Which year generated the highest sales?
2. Which months have the highest sales?
3. Which months have the lowest sales?
4. Is the business growing year over year?
5. Are there any unusual sales spikes or drops?
6. Which quarter performs best?
7. How does the current year's performance compare with the previous year?

11. Phase 7 – Product Performance Analysis

Create a **Product Performance Dashboard**.

Analyze:

- Sales by product
- Quantity sold by product
- Cost by product
- Profit by product
- Profit margin by product
- Average selling price
- Product contribution to total sales

Business Questions

1. Which products generate the highest revenue?
2. Which products generate the highest profit?
3. Which products have high sales but low margins?
4. Which products have the highest quantity sold?
5. Which products should management prioritize?

6. Which products may require pricing or cost optimization?

Recommended Visuals

- Top 10 Products
- Bottom 10 Products
- Sales vs Profit scatter plot
- Product contribution chart
- Product profitability table
- Product-wise sales trend

12. Phase 8 – Customer Analysis

Create a **Customer Insights** page.

Analyze:

- Total customers
- Sales by customer
- Orders by customer
- Quantity purchased
- Average order value
- Customer contribution %

Business Questions

1. Who are the top customers?
2. What percentage of revenue comes from the top 10 customers?
3. Which customers generate the highest order volume?
4. Which customers contribute the highest profit?
5. Are there customers with high sales but low profitability?
6. Which customers should the organization prioritize?

13. Phase 9 – Salesperson Performance

Create a **Salesperson Performance Dashboard**.

Analyze:

- Salesperson-wise sales
- Number of orders
- Quantity sold
- Profit generated
- Average order value
- Sales contribution %
- YoY performance

Business Questions

1. Who is the highest-performing salesperson?
2. Who generates the highest profit?
3. Who handles the highest number of orders?
4. Which salespeople are improving year over year?
5. Are there salespeople with high sales but low profitability?
6. Which salespeople may require additional support or training?

14. Phase 10 – Geographic Analysis

Use the **Locations** table to analyze geographical performance.

Analyze:

- Sales by city
- Sales by state
- Sales by location
- Sales vs population
- Sales vs household information
- Sales by geographic region

Business Questions

1. Which states generate the highest revenue?
2. Which locations are performing best?
3. Which locations have low sales?
4. Does population appear to influence sales?
5. Are there locations with high population but relatively low sales?
6. Which locations could represent potential growth opportunities?

Recommended Visuals

- Filled Map / Map
- State-wise sales
- Location ranking
- Sales vs Population scatter chart
- Location performance matrix

15. Phase 11 – Budget vs Actual Analysis

Use the **Budgeting** table to build a financial performance view.

Management wants to understand:

"Are we achieving our expected revenue?"

Analyze:

- Budget
- Actual Sales
- Variance
- Variance %
- Budget Achievement %

Business Questions

1. Which locations exceeded their budget?
2. Which locations failed to achieve their budget?
3. What is the overall budget achievement percentage?
4. Which months have the highest budget variance?

5. Which locations consistently underperform against budget?
6. Which locations should management investigate?

Recommended Visuals

- Actual vs Budget column chart
- Variance chart
- Budget Achievement KPI
- Location-wise budget matrix
- Monthly budget vs actual trend

16. Phase 12 – Interactive Dashboard Features

The dashboard should not be a collection of static charts.

Implement interactive Power BI features such as:

Slicers

At minimum:

- Year
- Month
- State
- Location
- Product
- Customer
- Salesperson

Drill Down

Allow users to navigate:

Year → Quarter → Month → Day

and where appropriate:

State → Location

Drill Through

Create drill-through pages for:

- Product
- Customer
- Salesperson
- Location

Tooltips

Create meaningful report tooltips containing additional business information.

Cross Filtering

Ensure that selecting one visual appropriately filters related visuals.

17. Advanced Power BI Requirements

Advanced DAX

Implement:

- CALCULATE()
- FILTER()
- ALL()
- ALLSELECTED()
- DIVIDE()
- SUMX()
- AVERAGEX()
- DISTINCTCOUNT()
- DATEADD()
- SAMEPERIODLASTYEAR()
- TOTALYTD()
- RANKX()

Advanced Analytics

Try to implement:

- Top N analysis
- Pareto analysis

- Dynamic ranking
- Dynamic titles
- YoY comparison
- Running totals
- Moving averages
- Contribution %
- Conditional formatting
- KPI indicators

18. Management Questions – Final Challenge

At the end of the project, students should be able to answer the following questions using their Power BI report.

Business Performance

1. What is the total revenue?
2. What is the total profit?
3. What is the overall profit margin?
4. Is revenue growing year over year?
5. Which year performed best?

Products

6. Which are the top 10 products by revenue?
7. Which are the top 10 products by profit?
8. Which products have poor margins?
9. Which products have the highest demand?

Customers

10. Who are the top 10 customers?
11. What percentage of sales comes from the top customers?
12. Which customers generate the highest profit?

Sales Team

13. Who is the best-performing salesperson?
14. Who generated the highest profit?
15. Which salesperson has the highest order volume?

Geography

16. Which state generates the highest revenue?
17. Which locations are underperforming?
18. Which locations appear to have growth potential?

Budget

19. Is the organization achieving its budget?
20. Which locations exceeded the budget?
21. Which locations missed the budget?
22. Which months had the largest budget variance?

Strategic Insights

23. What are the three biggest strengths of the business?
24. What are the three biggest areas of concern?
25. What recommendations would you give management based on your analysis?

19. Expected Dashboard Structure

Page 1 – Executive Overview

Overall KPIs and business performance.

Page 2 – Sales Trend Analysis

Year/month/quarter trends and YoY analysis.

Page 3 – Product Performance

Product sales, quantity, profit and margins.

Page 4 – Customer Analysis

Customer contribution and ranking.

Page 5 – Salesperson Performance

Individual salesperson performance.

Page 6 – Geographic Analysis

State/location performance and map analysis.

Page 7 – Budget vs Actual

Budget achievement and variance analysis.

20. Deliverables

Each student/team should submit:

1. Power BI Report

.pbix file containing:

- Clean data
- Data model
- Relationships
- DAX measures
- Calculated columns where required
- Interactive dashboards

2. Business Insights

Prepare a short document containing:

- Top 5 findings
- Top 5 business problems identified
- Top 5 recommendations

3. Presentation

You should present their dashboard and explain:

- Data preparation
- Data model
- DAX measures
- Dashboard design
- Key findings
- Business recommendations